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RESEARCH PAPER

Automatically Generating BPMN 2.0 Process Models from Natural Language Process Descriptions: Challenges, Framework, Quality Assessment

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Abstract The increasing importance of process models highlights the demand for tools that enable an efficient and accurate transformation of natural language process descriptions into BPMN 2.0 process models. Despite significant advancements in natural language processing and large language models, automatically generating process models that are syntactically correct, semantically accurate, and pragmatically useful remains a challenge, particularly for individuals without deeper expertise in process modeling. To tackle this challenge, the authors have been developing BPMNGen, an LLM-based conversational framework, which leverages advanced natural language processing and large language models to automate the generation of BPMN 2.0 process models from natural language process descriptions. To evaluate the quality of the models generated with BPMNGen, two studies were conducted: one study focused on the semantic quality of the generated process models in relation to given process descriptions and another one assessed process model comprehensibility, including the cognitive load of the study participants when reading the LLM-generated models, their level of acceptability, and their performance. The results show that LLMs already perform well for simpler and moderately complex processes, while struggling with higher complexity – highlighting both current limitations and opportunities for further improving automated BPMN 2.0 model generation.

Keywords LLM · Automated model generation · BPMN 2.0 · Process model quality

1 Introduction

Process modeling constitutes a fundamental task of modern organizational management, enabling businesses to document, visualize, analyze, and optimize their business processes to improve their efficiency and effectiveness (Alves et al. 2023). At its core, process modeling involves the representation of business processes in a well-defined visual notation, which, for example, covers the sequence of tasks, routing decisions, multi-party interactions, flow of data, and resources.

A widely adopted business process modeling language is BPMN 2.0, which constitutes an internationally recognized standard (Object Management Group 2011). As opposed to other process modeling languages such as UML activity diagrams or Event-Driven Process Chains, BPMN 2.0 provides a higher expressiveness, enabling the modeling of complex control flow patterns, collaborations, subprocesses, and choreographies. Moreover, the comprehensive set of graphical elements provided by BPMN 2.0 facilitates communication across various stakeholders (Object Management Group 2011).

However, transcribing a natural language process description into accurate and correct BPMN 2.0 process models as well as modifying these models are challenging tasks, particularly for users without profound expertise in process modeling (Zimoch et al. 2018). Recent advancements in artificial intelligence (AI), particularly in natural

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language processing (NLP) and large language models (LLMs), enable new opportunities to tackle this challenge. These technologies have shown promising capabilities in understanding and generating natural language with high contextual accuracy (Klievtsova et al. 2023a, 2024a). However, many existing works are only able to create simple flow charts (e.g., sequences with simple gateways) neglecting, for example, the resource and data perspectives, or collaboration aspects (e.g., message exchanges between multiple process parties). Moreover, many approaches lack a profound assessment of both the quality of the created models (i.e., syntactic, semantic and pragmatic quality) and the quality of the model-generation process. This article focuses on the quality of LLM-generated process models.

To harness the full potential of the automated generation of BPMN process models from natural language process descriptions, this article presents *BPMNGen*, our LLM-based conversational framework to foster the creation of BPMN 2.0 process models through a conversational interface. With BPMNGen, users can describe business processes in natural language and iteratively refine their process descriptions and the process models automatically generated from them.

Beyond the implementation of the framework, our objective is to systematically assess quality aspects, including the quality of the generated process models, as well as the quality of the model-generation process (Hörner 2025; Möller 2025). This article presents two empirical studies we conducted to assess the quality of the generated process models in a controlled experiment. The first study investigates the *semantic quality* of BPMNGen-generated process models by comparing them with expert-created process models regarding their alignment with predefined process descriptions. The second study assesses the *comprehensibility* of the automatically generated process models, considering the ability of the participants to comprehend the generated process models, measuring their cognitive load, as well as their level of acceptability and their performance. Our empirical evaluation confirms that BPMNGen is capable of generating process models with a semantic accuracy comparable to expert-created models. Furthermore, our user studies show that the automatically generated process models are generally easy to understand, and in certain cases, even outperform expert models in respect to comprehensibility. These findings indicate that LLM-based frameworks like BPMNGen can substantially support process modeling tasks even for more complex scenarios. On the other hand, several limitations need to be acknowledged: the evaluation focused on relatively simple scenarios and a non-expert participant group, and parts of the measurement relied on self-reports. Moreover, the fast-evolving nature of LLMs may affect reproducibility over time. With this article, we not only aim to demonstrate that LLM-based generation of process models is feasible in principle, but also emphasize that quality assessment is fundamental for ensuring the reliability, applicability, and acceptance of such automatically generated models.

The remainder of this article is structured as follows. Section 2 provides the fundamental background information necessary to understand this work. Section 3 discusses related work, followed by Sect. 4 which presents our overall vision. Section 5 elicits user requirements for BPMNGen and presents the design of the BPMNGen architecture. Section 6 presents the BPMNGen proof-of-concept and implementation details. Section 7 evaluates BPMNGen with two empirical studies. Section 8 discusses the limitations and the threats to validity. Finally, Sect. 9 concludes the article.

2 Backgrounds

This section provides a detailed overview of the foundational concepts and technologies employed in the engineering and evaluation of BPMNGen.

2.1 Large Language Models

LLMs utilize deep learning techniques to understand, interpret, and generate natural language. Models like ChatGPT OpenAI 2025 and Gemini Google 2025 are trained on vast textual corpora, enabling them to recognize patterns, context, and nuanced language use. These models are built on the Transformer architecture (Vaswani et al. 2017), which introduced a novel attention mechanism that allows for efficient and parallel processing of sequential data. Building on this foundation, recent research has demonstrated the applicability of LLMs in specialized domains such as conceptual modeling in general (Reinhartz-Berger et al. 2025) and process modeling in particular (Kourani et al. 2024a). Key characteristics of LLMs include their ability to generate human-like text, handle context, answer questions, and assist users in performing creative tasks (Brown et al. 2020; Radford et al. 2019). An essential factor in harnessing the full potential of LLMs is the concept of *prompting*, which refers to the design of input instructions to guide the behavior of the LLM and its responses. Prompting serves as a crucial bridge between the user's intent and the output of the LLM, enabling its effective application across a wide range of tasks. The structure, clarity, and contextual framing of a prompt play a significant role in shaping the accuracy, coherence, and relevance of the generated responses (Liu et al. 2021).

2.2 Chatbots

Chatbots, also known as conversational agents, are software applications designed to simulate human-like interactions with users via textual or auditory communication (Meshram et al. 2021). Their main objective is to enhance user experience by providing assistance, automating tasks, and facilitating access to information through natural language interactions. This technology leverages components of AI such as NLP, machine learning (ML), and speech recognition to understand, interpret, and respond to user inputs (McTear et al. 2016). Modern chatbots often employ ML algorithms, particularly deep learning, to enhance language understanding and generation. They are trained on large datasets of human dialogues to produce contextually appropriate and coherent replies (Vinyals and Le 2015). Despite advances, chatbots still face challenges such as understanding complex user intents, managing ambiguous queries, and maintaining engagement in extended conversations. Recent innovations address these issues by incorporating advanced NLP techniques and context-aware computing (Bordes and Weston 2016). In particular, transformer-based models such as GPT have shown promising results in generating more natural and relevant interactions (Singh and Namin 2025).

2.3 Business Process Models

A process model is a formalized representation of a business process, capturing the sequence of events, tasks and interactions required to achieve a specific outcome. Process models allow understanding, analyzing, and improving business processes, often including tasks, decision points, message flows, pools, resources (e.g., actors or systems), and the flow of information or materials. Moreover, they are visualized using notations like BPMN and UML activity diagrams (Dumas et al. 2018).

BPMN 2.0 is a recognized standard, formally specified as ISO/IEC 19510:2013, for business process modeling (Object Management Group 2011). It offers a comprehensive graphical notation to model complex processes and enables a clear communication across stakeholders. A process model consists of at least one start and end event, tasks, and gateways. Start events initiate the process with one outgoing sequence flow, while end events conclude it with at least one incoming flow. Tasks represent work requiring resources, time, inputs, and outputs. Gateways control the process flow, managing splits and joins. Exclusive gateways (XOR) represent a decision point in the process where the flow branches into mutually exclusive paths. Only one path is executed, depending on the evaluation of defined conditions. Parallel gateways (AND) enable simultaneous tasks, dividing and rejoining flows to ensure synchronization. Splits and merges must use gateways of the same type, ensuring logical integrity and consistent process execution.

Fig. 1 shows an example of a BPMN 2.0 model depicting a typical supplier-seller-customer process for order fulfillment. It is noteworthy that the current version of BPMNGen is already able to create such multi-party process models.

3 Related Work

The increasing integration of LLMs into the field of Business Process Management (BPM) has led to a surge of approaches aiming to translate natural language descriptions into formal process models, particularly using BPMN 2.0 (Nour Eldin et al. 2024). Recent approaches increasingly leverage conversational or generative LLMs such as GPT-3.5, GPT-4, or open-source alternatives like LLaMA 3 to support process modeling tasks (Klievtsova et al. 2023b; Neuberger et al. 2024; Reinhartz-Berger et al. 2025). Several frameworks have emerged that embed LLMs in tools or workflows for process model generation. For example, *Nala2BPMN* employed a modular architecture combining OpenAI's GPT with LangChain to automatically extract, refine and structure process elements from text descriptions (Nour Eldin et al. 2024). *Nala2BPMN* has been evaluated using expert ratings on metrics such as correctness, completeness, and ease of updating. Similarly, *ProMoAI* and another work focused on zero-shot prompting strategies to transform natural language into BPMN process models, often evaluated through user feedback or output refinement cycles (Kourani et al. 2024b; Köpke and Safan 2024).

A number of studies centered on prompt engineering and input optimization. Köpke and Safan (2024) introduced cost-efficient prompting mechanisms and showed that performance can vary significantly depending on instruction design. Others, such as Neuberger et al. (2024) developed universal prompting strategies to generalize across domains. Still, the overall semantic fidelity of LLM-generated process models remains an open issue, as discussed by Fettke and Houy (2025) as well as Fill et al. (2023), who critically assessed conceptual modeling tasks handled by LLMs and found substantial variation in model accuracy and structure.

Conversational agents like *BPLLM* (Bernardi et al. 2024) and *LLM4PM* (Ziche and Apruzzese 2024) extended the paradigm by enabling users to interactively refine process models through dialogue, supporting exploratory and corrective iterations. These systems were evaluated with respect to answer quality, structural validity, and dialogue consistency, indicating progress towards real-

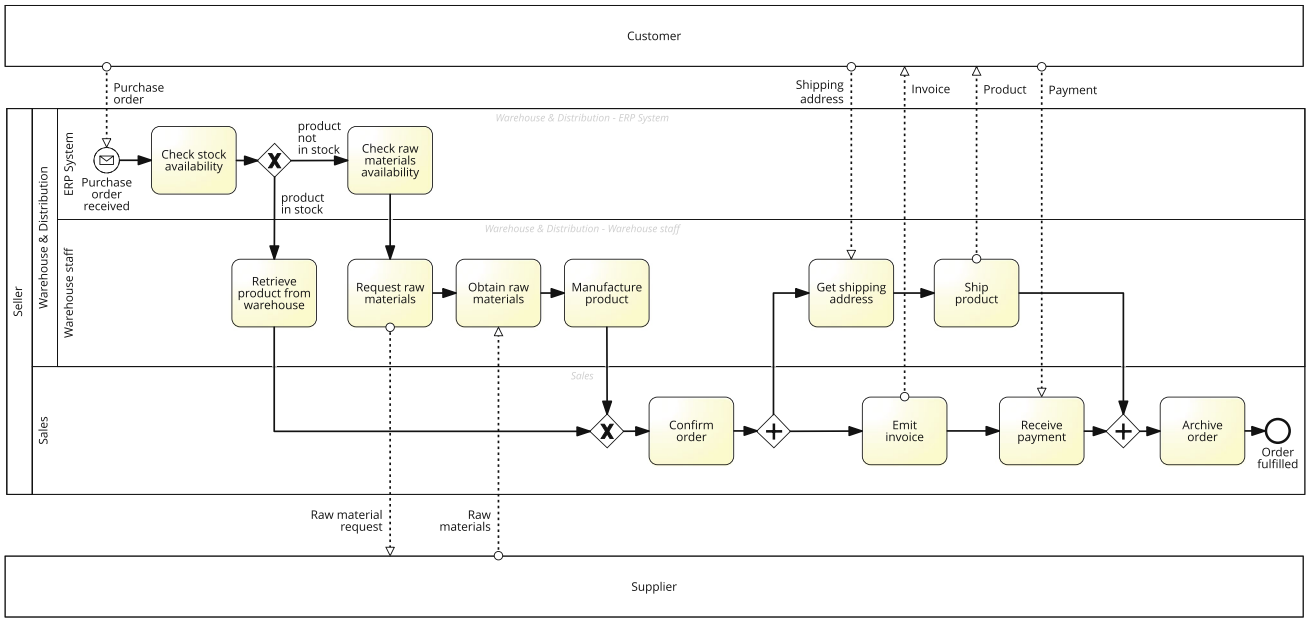


Fig. 1 Example of a process model in BPMN 2.0 (Dumas et al. 2018)

world applicability. Klievtsova et al. (2023b) explored the role of ChatGPT as a modeling assistant in educational scenarios and found that even simple prompts can lead to structurally consistent models, but often lack semantic depth. Similarly, Klievtsova et al. (2024c) examined how generative AI can support domain experts in modeling tasks and emphasized the need for iterative refinement strategies and appropriate prompt scaffolding.

Beyond generative tools, Grohs et al. (2023) provided a broader perspective on the applicability of LLMs in BPM, and identified key challenges such as hallucinations and the lack of formal constraints. At the same time, Monti et al. (2024) and Klievtsova et al. (2024b) addressed how natural language descriptions can be transformed into process fragments or activity patterns, particularly for their integration with existing models.

Some approaches remain grounded in traditional architectures. For instance, Ajmal et al. (2024) applied pipeline-based parsing and semantic role labeling, and showed competitive precision and recall, even without LLMs. Likewise, rule-based extraction methods (e.g., Fill et al. (2023)) remain relevant for highly structured domains or where explainability is critical.

In addition to recent LLM-based solutions, earlier research explored BPMN model generation from text using non-generative approaches. For example, Friedrich et al. (2011) introduced a structured method for generating BPMN models based on syntactic parsing and transformation rules. Ivanchikj et al. (2020) proposed a system that supports end-users in translating text into BPMN models, and evaluated the usability and accuracy of the created models. Bellan et al. (2021) provided a benchmark of text-to-process-model techniques and outlined key research challenges in this domain. These works constitutes an important foundation and complement current efforts by highlighting longstanding challenges in semantic interpretation, user involvement, and system evaluation.

Building an operational chatbot that generates BPMN 2.0 process models from natural language is a non-trivial task. Although recent advances in LLMs have demonstrated the general feasibility of automated process modeling, most existing approaches stop at conceptual prototypes or isolated experiments. Concrete implementations – particularly those that integrate prompt handling, modeling languages, and user interaction – are rare, and practical design considerations are often left unaddressed. Finally, many approaches restrict themselves to rather simple model behavior (e.g., sequences with simple gateways).

While prior approaches demonstrate the feasibility of generating BPMN models from natural language, they typically focus on prompting strategies, structural aspects, or informal expert judgments. What distinguishes our work is the systematic evaluation of LLM-generated process models across the two key quality dimensions: semantic alignment and pragmatic quality, reflected in the comprehensibility of the process models. Through controlled user studies comparing BPMNGen-generated process models with expert-created counterparts, we provide empirical evidence on how well end users understand, interpret, and accept these models. This focus on human-centered evaluation addresses a fundamental aspect in automated

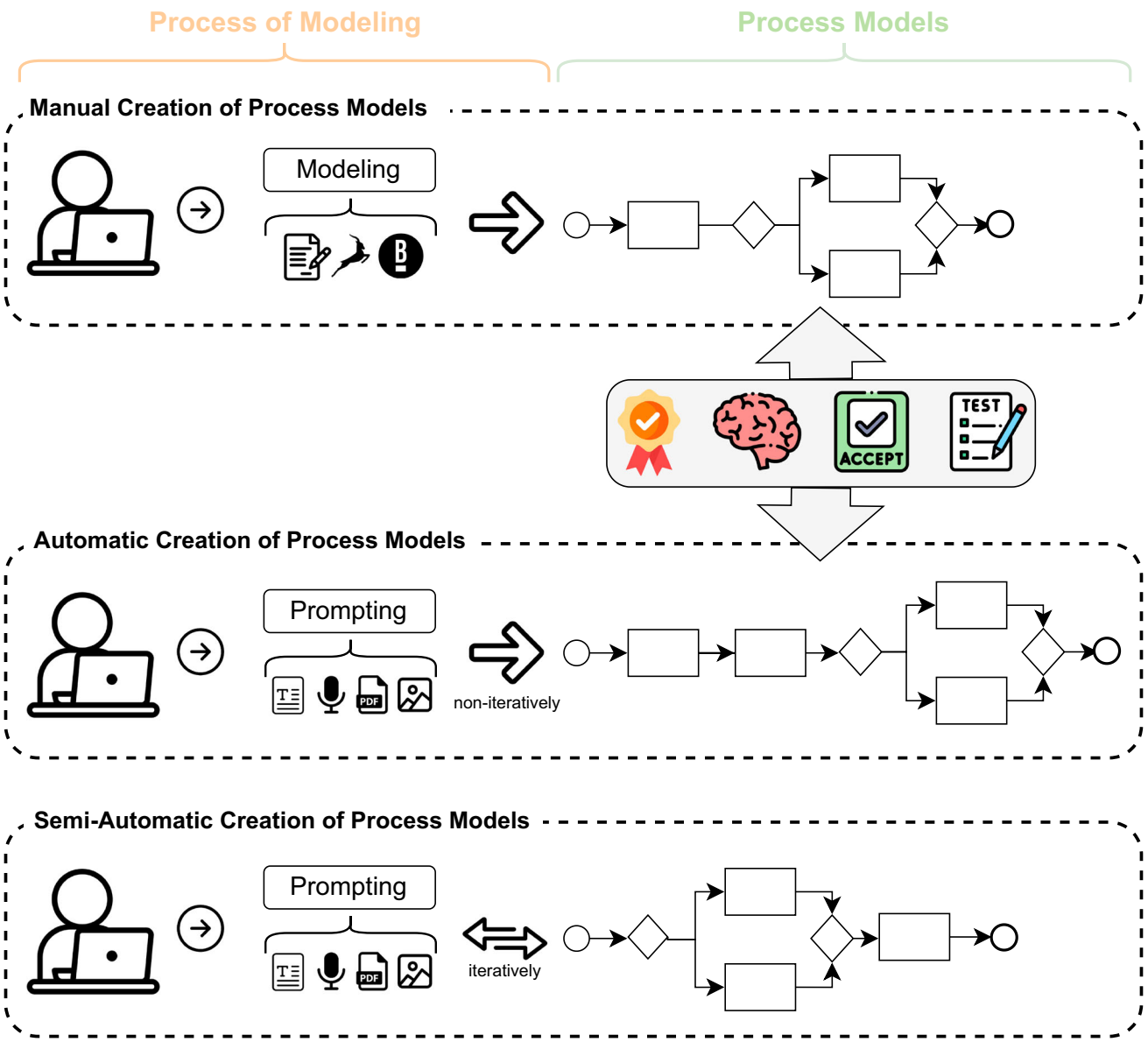


Fig. 2 Vision

process model generation that has so far remained largely unexplored in the current state of the art.

4 Vision

To enhance the state of the art in LLM-based process model generation, this article systematically compares two modeling paradigms: (1) BPMN 2.0 process models manually created by a domain expert, and (2) BPMN 2.0 process models automatically generated by BPMNGen (i.e., the LLM-based conversational framework). In our setting, "automatically generated" refers to a zero-shot prompting approach in which the complete textual process description is provided to BPMNGen and a BPMN 2.0 model is generated in a single step without iterative user interaction.

In addition to these two paradigms, a third category of process models becomes relevant: (3) semi-automatically generated process models, which are created through an iterative collaboration between the user and the LLM. In this interactive mode, users refine or extend the model through multiple prompts (e.g., step-by-step instructions or

chain-of-thought style guidance). While such semi-automatic modeling constitutes an important direction for future research, it is not the focus of this article (cf. Fig. 2).

We aim to evaluate the resulting models along key dimensions such as semantic quality, cognitive load, level of acceptability, and comprehension performance. This allows us to better understand how well LLMs perform in practical modeling scenarios – not only in terms of what they generate, but also how usable and understandable their output is to humans.

5 Preparation Phase

This section outlines the key steps taken in the preparation phase prior to the engineering of the LLM-based conversational framework BPMNGen. These key steps are essential for guiding the engineering process, ensuring that the resulting framework meets the goals of the project and end-user requirements.

5.1 Eliciting User Requirements

The user requirements were initially identified through brainstorming sessions involving representative stakeholders with relevant domain knowledge and practical process modeling experience. To ensure a transparent and systematic elicitation process, we conducted a structured brainstorming session with six stakeholders in total: three BPM experts from academia and three experts from a process management tool vendor (i.e., *AristaFlow BPM Suite*, Dadam et al. (2009)). All participants were familiar with both BPMN 2.0 and LLM-based technologies.

The brainstorming followed a structured elicitation approach commonly used in requirements engineering. First, all stakeholders independently documented potential user requirements. In a subsequent moderated consolidation step, these inputs were jointly reviewed, merged, and clarified to resolve overlaps or ambiguities and to ensure shared understanding across participants. This procedure fostered balanced contributions and prevented dominance effects.

Following the consolidation of requirements, we applied the MoSCoW method (Kravchenko et al. 2022) to prioritize them based on their relative importance. MoSCoW is a widely used technique in requirements engineering that classifies requirements into four categories: 'Must have' (essential features critical for the success of BPMNGen), 'Should have' (important but not critical and possibly deferrable), 'Could have' (nice-to-have features that add value but are not essential), and 'Won't have' (out of scope for the current iteration). In the requirements elicitation process, we intentionally chose not to collect any 'Won't have' requirements. This decision was made as our primary focus was on the essential and valuable features that directly contribute to the success of BPMNGen.

The requirements were then categorized into functional and non-functional groups. This categorization is a common practice in requirements engineering to clearly differentiate between system capabilities and performance attributes. Functional requirements define the specific actions BPMNGen needs to offer to meet user needs, focusing on features that directly impact the user experience. Non-functional requirements, which will be discussed separately, relate to the performance, usability, and reliability of the framework. The gathered user requirements are listed in Tables 1 and 2.

5.2 BPMNGen Architecture

Based on the preparation phase, the BPMNGen architecture follows a client–server model to enable scalability and concurrent user access. The architecture (cf. Fig. 3) is built around four primary components: custom assistant, server, application, and database. These components collaborate to enable the automated creation, visualization, and management of BPMN 2.0 process models. The user interacts with the application by providing instructions and performing tasks such as editing and reviewing.

Custom Assistant. The custom assistant is powered by the LLM provider (i.e., OpenAI in the given article) and accessed via its API. It processes user-provided instructions, transforming them into a JSON structure that contains all the information necessary to generate a BPMN 2.0 process model. This component acts as the core of BPMNGen for NLP and intelligent interpretation. We use *GPT-4o*[1] as the LLM, as it was widely regarded as one of the most capable and advanced models at the time of implementation, particularly in terms of natural language understanding and contextual accuracy. Its strong performance across various NLP tasks made it a suitable choice for our requirements.

Server. The server acts as the central orchestrator, connecting all other components. It forwards user instructions to the custom assistant and receives the JSON output. The server is also responsible for converting this JSON output into an XML document, which is then used to define the BPMN 2.0 process model. Additionally, it manages access to the database where model files are stored, ensuring data consistency and reliability.

Application. The application serves as the user interface and enables interactions between the user and BPMNGen. It provides an input panel for instructions, a process model view, and tools for editing BPMN 2.0 process models. The

[1] We will always integrate the latest available GPT version.

Table 1 Functional requirements for BPMNGen

1. Must Haves	
Process Model Creation (Text-to-Model)	The LLM-based conversational framework must allow users to input natural language descriptions to create BPMN 2.0 diagrams
Process Model Visualization	The generated BPMN 2.0 process model must be displayed to the users
Process Model Modification	The users must be able to modify the created process models directly through the LLM-based conversational framework. This includes editing process models either by entering a new update prompt or by manually interacting with the visual diagram using drag-and-drop
2. Should Haves	
Generated Process Models History	The LLM-based conversational framework should allow users to access and view all generated BPMN 2.0 models. This ensures that users can easily review and manage previously created process models directly from their conversation history
Process Model Saving	Users should be able to store the process models they created, such that they can retrieve and further edit them later
Process Model Export/ Download	The users should be able to export the generated process model in various formats (e.g., PDF, BPMN)
Save Chat History with Data	The LLM-based conversational framework should store the complete chat history associated with each process model, including all relevant information such as the name of the process model, time of creation/last modification, user prompts, and the process model itself
Error Feedback	If an error occurs in the backend that could disrupt a process in the frontend, the users should be alerted
3. Could Haves	
Natural Language Creation (Model-to-Text)	The LLM-based conversational framework could allow users to provide a BPMN 2.0 diagram as input and to create a detailed natural language process description as output. This description should outline the elements and context of the process in natural language for better understanding and documentation purposes
Collaboration	The users could be able to share his generated process models with other users and collaborate on them, so that they can work together efficiently and exchange knowledge
Help and Documentation	The LLM-based conversational framework could provide clear help and documentation or suggestive UI design to guide users through the process of using the LLM-based conversational framework
Multi-Language Support	The users could be able to interact with the LLM-based conversational framework in multiple languages

Table 2 Non-functional requirements for BPMNGen

1. Must Haves	
Access Control	The administrator must be able to manage access to the process models and editing rights, so that only authorized persons may create models or change them
2. Should Haves	
Usability	The LLM-based conversational framework should provide an intuitive and user-friendly interface. It should be consistent and easy to navigate. Error messages should be clear, so the users know what to do or what has gone wrong
Reliability	Data integrity should be ensured, without loss of critical information. Hence, the LLM-based conversational framework should run consistently without failure
Performance	In the interest of user experience, the LLM-based conversational framework should perform its functions within a reasonable time span
Availability	The LLM-based conversational framework should be accessible and operational for the users whenever needed
3. Could Haves	
Accessibility	The LLM-based conversational framework could run on all commonly used internet browser clients, such as Chromium, Firefox or Opera

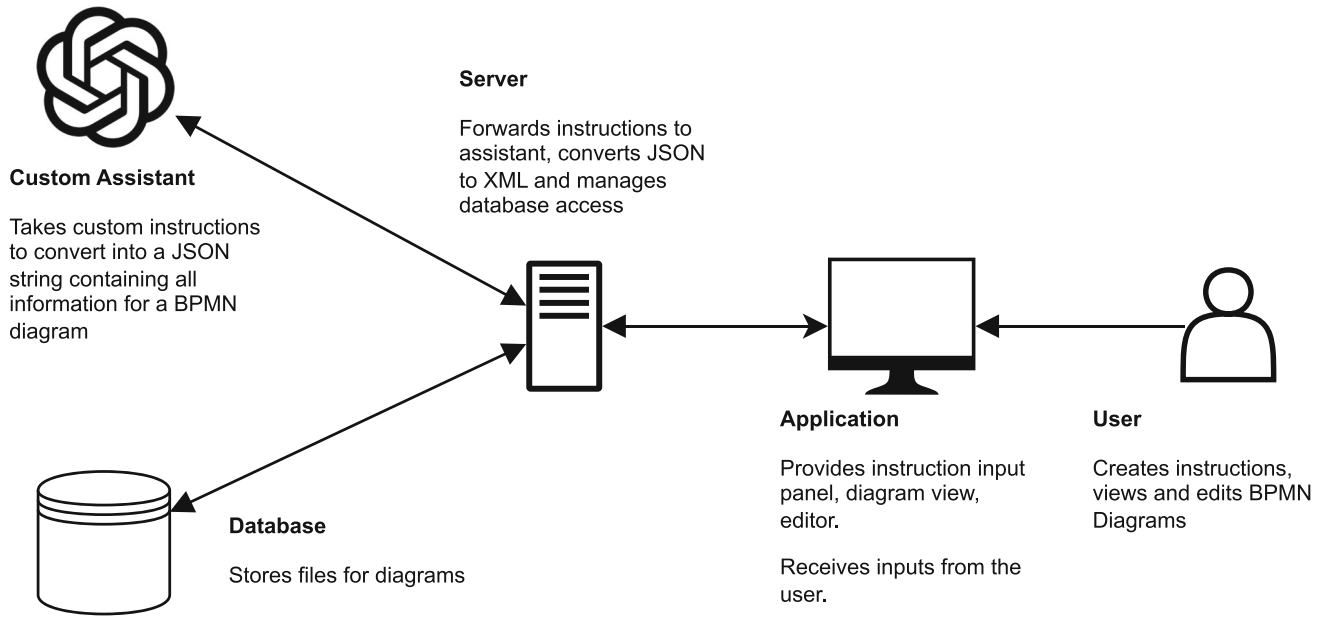


Fig. 3 Architecture of BPMNGen

application retrieves data from the server and ensures that users can easily create, view, and edit process models through an intuitive interface.

Database. The database is a key part of the architecture, storing all data required to generate, modify, and manage the BPMN 2.0 process models. This ensures that the process models are accessible for future use, iterative refinement, and analysis. As a pragmatic initial solution for the early development phase of BPMNGen, we implemented a lightweight, document-oriented data storage approach. Instead of relying on a full-fledged database management system, the current prototype utilizes locally stored, structured JSON files. The stored data is organized into three main components: threads, messages, and process models.

6 BPMNGen Implementation

This section outlines the engineering and implementation of BPMNGen, emphasizing how BPMN 2.0 business process models can be generated from textual process descriptions. Section 6.1 presents the BPMNGen frontend, while Sect. 6.2 gives insights into the BPMNGen backend, detailing the various components and their roles in the overall architecture. Section 6.3 revisits the functional and non-functional requirements for BPMNGen, categorizing them based on their implementation status.

6.1 BPMNGen Frontend

BPMNGen enables users to create and iteratively refine BPMN 2.0 process models using natural language input. As an example of a BPMNGen use case, consider a scenario in which a user provides a process description in natural language, which is then converted into an initial BPMN 2.0 process model displayed in the central workspace (see Fig. 4 [A]). On the right side, the prompt history panel allows users to update the process model dynamically by entering additional instructions – such as *Add a notification step if the stock is unavailable* (see Fig. 4 [B]). The left panel gives access to previously generated models, allowing users to revisit and edit earlier process model versions (see Fig. 4 [C]) if needed. Once the user is satisfied with the refined process model, it can be saved, exported, or deleted using the top menu bar (see Fig. 4 [D]). This interactive workflow makes process modeling more accessible and intuitive through natural language interaction.[2]

6.2 BPMNGen Backend

To provide a structured overview of its functionality, the BPMNGen backend is examined through its two primary use cases: the creation of new process models and the modification of existing ones.

[2] A demonstration screencast of BPMNGen, as well as the complete source code, is accessible via the following link: https://anonymous.4open.science/r/BISE_Study_Materials-615C.

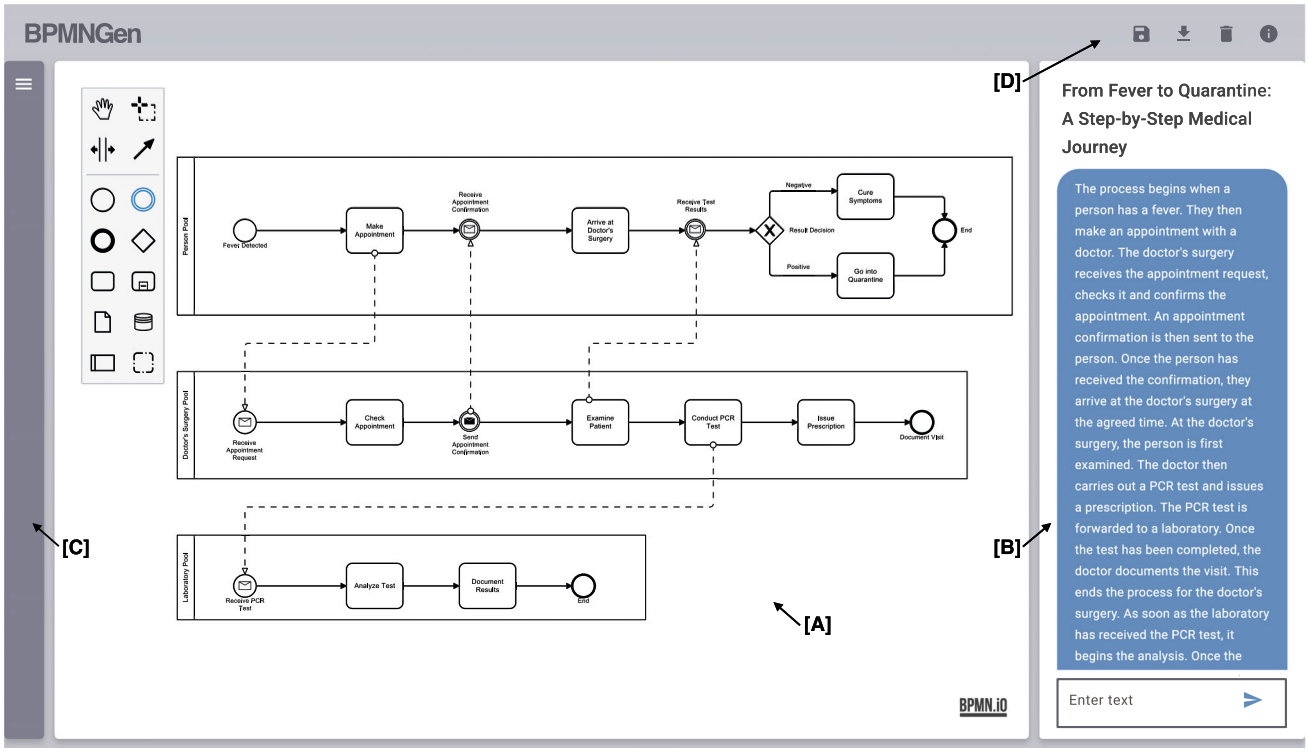


Fig. 4 Overview of the BPMNGen UI and a generated process model

6.2.1 *Process Model Creation*

The creation process starts when a user submits an initial natural language description of a business process through the application. This prompt is forwarded to the backend server, which initiates communication with the OpenAI API. To initiate an interaction, a *ChatCompletion* object is created by specifying several key parameters, including *messages*, *model*, and *temperature*. The *messages* parameter contains the input of the user, while the *model* parameter specifies the GPT model to be used. The *temperature* parameter controls the variability of the responses: lower values lead to more deterministic and reproducible outputs, while higher values allow for more diverse or creative responses. For BPMNGen, we select a low *temperature* to ensure consistent results across multiple iterations of the same prompt. In the implementation, we conducted preliminary tests with different *temperature* values and found that higher *temperatures* (e.g., > 0.2) tended to produce less consistent and sometimes hallucinated results, particularly when generating structured BPMN elements. Based on these observations, we opted for a *temperature* of 0.0 to ensure deterministic and stable outputs in the context of our use case. Additional parameters can be configured to further refine the response generation. For example, a maximum token limit can be set to constrain the length of the response, or a top-p value can be adjusted to balance creativity and coherence. Once the *ChatCompletion* object is instantiated, the procedure starts and the chat response is generated. To specialize the behavior for process modeling, BPMNGen uses the OpenAI custom assistant, which corresponds to a configurable version of its LLM, designed for specialized tasks within specific domains. As opposed to general-purpose models, the custom assistant can be adapted to particular applications by setting behavior rules, providing domain-specific knowledge, and using example-driven prompts (Lovin 2025). In the context of BPMNGen, the custom assistant is used to process user input and to generate structured responses for business process modeling. The interaction begins with retrieving the custom assistant via its unique ID and initializing a new thread to isolate the context for each user input. The input from the user is appended to this thread, and a function is initiated to trigger the assistant's response generation.

Upon completion of the function, the assistant retrieves the latest message, which is expected to contain a valid JSON structure. If the response is valid, the JSON data is stored in a file and added to the database together with the user prompt and thread ID. This ensures traceability and allows for further processing. Finally, the JSON file is converted to a BPMN 2.0 XML representation for frontend visualization and editing. This conversion consists of four steps (cf. Fig. 5): (1) reading and parsing the JSON input,

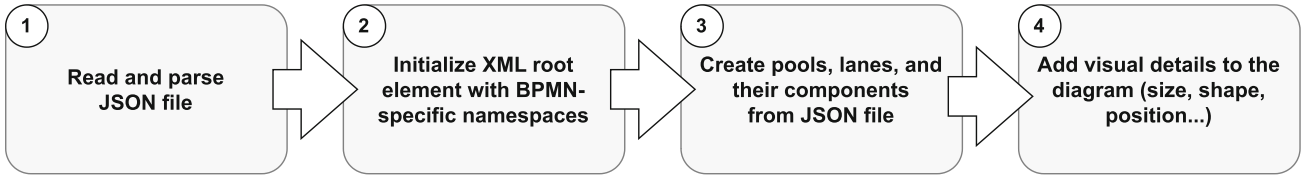


Fig. 5 Steps for converting JSON to XML

(2) initializing the XML structure with the required BPMN namespaces, (3) instantiating the BPMN elements, and (4) embedding the visual layout information (e.g., element positions, sizes, and flow paths). The layout information is directly taken from the LLM-generated JSON output and transferred into the XML structure, with only minor adjustments applied where necessary (e.g., adapting pool dimensions or refining flow connections). After completing the conversion, the resulting process model is rendered in the user interface of the application, where it can be visually inspected and iteratively modified by the user.

6.2.2 *Process Model Modification*

As opposed to model creation, modifying a model involves reloading and editing an existing file. The workflow begins when the user opens the application, which then sends a request to the server for a list of all available model entries. These model entries are retrieved from the local JSON-based storage and displayed in the application interface. Once a specific model is selected, the application requests the associated thread and diagram content from the server, which is then loaded into the editing interface.

To modify a process model, the user submits a new natural language instruction describing the intended changes. The server processes this instruction by initiating a new thread or appending to an existing one and sending the message to the custom assistant. As in the creation case, the assistant generates a structured JSON update message, which contains commands that indicate how the existing process model shall be changed (cf. Table 3).

The backend then uses an `updateJSON()` function to apply these commands to the original JSON structure of the model. Unlike full regeneration, this incremental update avoids redundant processing, reduces token usage, and allows for more efficient modifications. The updated JSON is then converted into an XML representation using the same transformation pipeline as in the creation process.

Before storing the updated XML file, it passes a dedicated validation process. Specifically, we implemented an XML validator that automatically analyzes and restructures the converted XML file where necessary. This validation is handled by a function called `validateXML()`, which converts the XML to JSON, applies structural corrections, and returns a fully compliant BPMN 2.0 XML document. As part of this process, the validator identifies message flows that are placed incorrectly, extracts them, and repositions them to the correct structural level to ensure compliance with the BPMN specification. Once validated, the XML file is stored, and the database is updated with the latest user instruction and a new timestamp. The application then refreshes the view on the process model.

6.3 Requirements Assessment

This section revisits the requirements set out for BPMNGen to determine the current state of their fulfillment (cf. Sect. 5.1). The requirements are re-listed in Tables 4 and 5, and categorized according to their current status (✓: Fully implemented and functioning as intended; ∼: Partially implemented or functional, but requiring further improvement to meet the desired standards; ✗: Not yet implemented or supported). The implementation status of the requirements was subsequently reviewed and validated by three BPM experts from academia and three experts from the process management tool vendor *Arista-Flow BPM Suite*.

Table 3 Command descriptions for diagram modification

Command	Description
CREATE	Adds new items (Pools, Lanes, Flows, or Components) to the diagram
ALTER	Modifies properties of existing items based on provided changes
DELETE	Removes an item (Pool, Lane, Flow, or Component) from the existing structure. In case of deleting a parent item, all children will be removed as well
UPDATEPARENT	Changes the parent of an item, effectively moving it to a different part of the structure

Table 4 Realized functional requirements for BPMNGen

1. Must Haves	
Process Model Creation (Text-to-Model)	✓
Process Model Visualization	✓
Process Model Modification	✓
2. Should Haves	
Generated Process Models History	✓
Process Model Saving	✓
Process Model Export/Download	✓
Save Chat History with Data	~
Error Feedback	✓
3. Could Haves	
Natural Language Creation (Model-to-Text)	✗
Collaboration	✗
Help and Documentation	✗
Multi-Language Support	✓

Table 5 Realized non-functional requirements for BPMNGen

1. Must Haves	
Access Control	✓
2. Should Haves	
Usability	✓
Reliability	~
Performance	~
Availability	✓
3. Could Haves	
Accessibility	✗

The evaluation of the functional and non-functional requirements of BPMNGen (cf. Tables 4 and 5) shows a solid foundation. Key 'Must have' requirements, such as process model creation, visualization, and modification, have been fully implemented, demonstrating the core capabilities of BPMNGen. However, some 'Should have' and 'Could have' requirements, such as saving chat history with associated data or offering collaboration features, either have been partially realized or are still under development. Furthermore, implementing 'Could have' features like model-to-text generation and improved accessibility are subject of future work. Such features could significantly enhance the versatility and appeal to a broader audience.

7 Evaluation

In addition to the provided proof-of-concept of BPMNGen, a key objective of our work is to thoroughly evaluate the quality of the automatically generated process models. To guide this evaluation, we first defined four specific research questions that deal with the semantic quality (including the soundness of the process models) and comprehensibility of the generated models. We designed two empirical studies addressing these research questions: Study I focused on evaluating the semantic quality, whereas Study II examined the comprehensibility of the automatically generated models. For each study, the remainder of this section presents the study design, the participants involved, the materials used, the performance measurements applied, the study results, and a discussion of the results.[3]

7.1 Research Questions

To systematically assess these quality aspects, we formulated a set of research questions grounded in complementary foundations. First, insights from related work highlight a pressing need to evaluate both the quality and comprehensibility of process models – particularly those generated by LLMs (Hörner and Reichert 2024; Möller et al. 2025). Second, we drew on established theoretical frameworks to guide our investigation. The *SEQUAL framework* (Krogstie 2016) conceptualizes semantic quality as the correspondence between a process model and the real-world domain it represents. In our setting, however, the underlying domain is provided through textual process descriptions, which inherently contain ambiguities typical for natural language. Accordingly, our study operationalizes semantic quality as the degree to which a generated process model aligns with the meaning that users derive from the textual description of the domain, which led us to our first research question:

RQ1: How does the semantic accuracy of process models automatically generated by BPMNGen compare to human expert-created process models across varying levels of process complexity?

To evaluate the mental effort involved in understanding these process models, we adopted the *Cognitive Load Theory*, which directly informed our second research question:

RQ2: How does the cognitive load experienced during the comprehension of process models automatically generated by BPMNGen compare to human expert-created process models across varying levels of process complexity?

[3] All study materials are available at: https://anonymous.4open.science/r/BISE_Study_Materials-615C.

Furthermore, the *Technology Acceptance Model* guided the formulation of our third research question, focusing on user perceptions of usefulness and acceptability:

RQ3: How does the level of acceptability of process models automatically generated by BPMNGen compare to human expert-created process models during comprehension across varying levels of process complexity?

Finally, we formulated a fourth research question that integrates perspectives from both process model quality and user comprehension:

RQ4: How does the comprehension performance of process models automatically generated by BPMNGen compare to human expert-created process models across varying levels of process complexity?

7.2 Study I: Semantic Quality of Generated Process Models

Study I, which addresses RQ1, is described in detail, and its results are discussed in Sect. 7.2.6.

7.2.1 Study Design

Study I consists of several steps. After providing informed consent, participants completed a demographic questionnaire. Then, a process description with two corresponding models was shown to them – one generated automatically by the presented LLM-based conversational framework (i.e., BPMNGen), the other one created manually by a human expert using the process modeling tool SAP Signavio. Each participant assessed the two models and selected the one that best represents the process description (cf. Fig. 6 for an example). This task was repeated across five model pairs. The study concluded after completing all model assessments.

7.2.2 Study Participants

Participation was entirely voluntary, and participants were assured that they could withdraw from the study at any time. Prior to their involvement, participants provided their consent electronically, confirming that they had been thoroughly briefed about the purpose, methodology, and goals of the study. The survey link was primarily distributed through university courses to recruit participants.

In Study I, a total of $N = 67$ participants were involved. All participants were informed in advance about the goals and procedures of the study. 27 participants were female and 40 male. The average age was 24.9 (median = 24.0). 61 participants stated that they were already experienced with BPMN 2.0 (No Experience = 6, Novice = 27, Intermediate = 19, Advanced = 13, Expert = 2). Table 6 summarizes this information.

7.2.3 Materials

Each of the two studies (i.e., Study I and II) was based on 5 distinct process scenarios. For each scenario, two BPMN 2.0 process models were created: one model was automatically generated using BPMNGen (with GPT-3.5 Turbo as the underlying LLM) and a second model was manually modeled by a human expert. This resulted in a total of 10 process models per study. All automatically generated models were produced in a controlled setup by the authors. Each process description was provided to BPMNGen as a complete natural-language description and processed using a zero-shot prompting approach. To ensure a comparable layout between the process models, we only adjusted poorly arranged sequence flows in the BPMNGen-generated process models. These changes were strictly visual and limited to improving the placement of sequence flows that were overly bent or awkwardly positioned. Semantic elements, model structure, or modeling content were not modified. The expert-created process models were modeled directly in SAP Signavio without any LLM support. To ensure a consistent visual style between these manually created models and the automatically generated ones, all BPMNGen output models were redrawn in SAP Signavio as well. This approach ensured that both model sets shared the same layout conventions, element colors, and general visual appearance, preventing participants from recognizing tool-specific layout artifacts. The 5 process models (PM) covered the following scenarios: 1) online shopping; 2) baking pizza; 3) borrowing a book; 4) making a coffee; 5) requesting vacation.

On average, BPMNGen-generated process models contain 12 modeling elements, whereas the process models created by the expert with SAP Signavio contained an average number of approximately 14 elements. Meta-data about the process models – including the number of elements per type for both expert and BPMNGen models – is shown in Table 7. To enhance the complexity of the process models, increasingly detailed process descriptions were employed, with the description of PM1 being the simplest one and the description of PM5 being the most sophisticated one.

7.2.4 Performance Measure

In the following, we provide a detailed description of the performance measurement used in Study I.

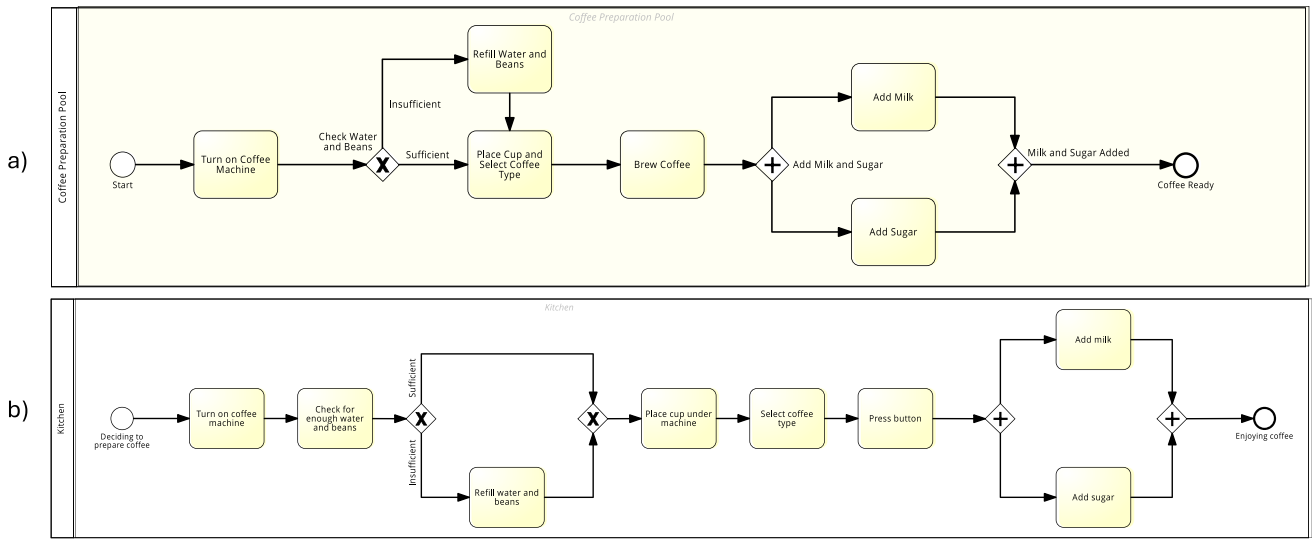


Fig. 6 Extract from Study I – Process description and corresponding process models: (a) represents BPMNGen-generated model, while (b) depicts the model generated by an expert

Table 6 Sample description of study I

Variable	Participants ($N = 67$)
Gender N (%)	
female	27 (40.3)
male	40 (59.7)
Age N (%)	
$<$ 25 years	36 (53.7)
$\geq$ 25 years	31 (46.3)
Education Level N (%)	
High school diploma or equivalent	16 (23.9)
Bachelor Degree	39 (58.2)
Master Degree	10 (14.9)
Doctor of Philosophy (PhD)	2 (3.0)
Experience N (%)	
no experience	6 (9.0)
novice	27 (40.3)
intermediate	19 (28.4)
advanced	13 (19.4)
expert	2 (3.0)



Fig. 7 Research model for study I

Table 8 Descriptive results for RQ1

PM	BPMNGen		Expert	
	Frequency	Percentage	Frequency	Percentage
PM1	**39**	58.2	28	41.8
PM2	**34**	50.7	33	49.3
PM3	30	44.8	**37**	55.2
PM4	21	31.3	**46**	68.7
PM5	19	28.4	**48**	71.6

Table 7 Number of modeling elements for each process model

PM	Expert						BPMNGen					
	Act.	Evt.	Gtw.	Pool	Lane	Total	Act.	Evt.	Gtw.	Pool	Lane	Total
PM1	5	2	0	1	1	**9**	6	2	0	1	1	**10**
PM2	7	2	1	1	1	**12**	7	2	1	1	1	**12**
PM3	8	2	4	1	1	**16**	4	2	2	1	1	**10**
PM4	8	2	4	1	1	**16**	6	2	3	1	1	**13**
PM5	9	3	3	1	2	**18**	6	3	3	1	2	**15**

7.2.4.1 Semantic Quality

In Study I (RQ1), we deal with semantic model quality as defined by the SEQUAL Framework (Krogstie 2016). Semantic quality refers to the extent to which a process model accurately represents the intended meaning of the natural language input. Semantic quality reflects the ability of the model to precisely and comprehensively capture the concepts, relationships, and sequences conveyed, ensuring semantic alignment with the real-world process. Fig. 7 illustrates the research model for Study I. The theoretical factor (F) is process model creation by either a chatbot or an expert. The operationalization of the factor (O) is semantic quality, measured through accuracy.

7.2.5 Results of Study I

This subsection presents the descriptive as well as inferential statistics of the results we obtained in the context of Study I.

7.2.5.1 Descriptive Statistics

Table 8 summarizes the descriptive results of Study I, highlighting the frequency distribution and corresponding percentage values of the participants' choices. The frequency indicates how many participants rated either the 'BPMNGen' or the 'Expert' process model as being better suited to represent the displayed process description, while the percentage provides a relative measure of these preferences.

The results depicted in Table 8 indicate that the participants' preferences vary significantly across the five process models (PM1-PM5). For PM1 and PM2, the 'BPMNGen' model was favored, with 58.2% and 50.7%. Conversely, for PM3 to PM5, the process models created by the expert were preferred, with a particularly strong majority in PM4 (68.7%) and PM5 (71.6 %).

7.2.5.2 Inferential Statistics

To determine whether a particular model was selected significantly more often than another, and to answer *RQ1* statistical analyses on the observed data were conducted.

For variables with two categorical outcomes, a binomial test was applied to evaluate whether the observed frequencies significantly deviated from an equal distribution ($p = 0.5$). This test indicated whether participants showed a significant preference for one of the two models (i.e., the model either created manually by the expert or automatically by BPMNGen). Observed frequencies and p-values were used to assess the statistical significance of these preferences, with conclusions drawn based on a conventional significance level of 0.05. The results indicate no significant preference between the models generated by BPMNGen and by the expert for PM1 (p = 0.222), PM2 (p = 1.000), and PM3 (p = 0.464). However, for PM4 (p = 0.003) and PM5 ($p < 0.001$), a significant preference for the models created by the expert was observed (cf. Table 9).

Table 9 Inferential statistics for RQ1

PM	BPMNGen (N)	Expert (N)	p-value (2-tailed)
PM1	39	28	0.222
PM2	34	33	1.000
PM3	30	37	0.464
PM4	21	46	0.003
PM5	19	48	<0.001

7.2.6 Discussion

Study I focused on the semantic quality (RQ1) of BPMNGen-generated vs. expert-created process models, with the study participants being tasked to evaluate which model was better aligned with the respective process description. The findings revealed notable variations in the participants' preferences across the five process models PM1–PM5.

For simpler process scenarios such as PM1 and PM2, BPMNGen-generated models were not only rated comparably to those created by the expert but even showed a slight preference in the case of PM1. Importantly, no significant differences were observed for PM1, PM2, and PM3, indicating that participants did not consistently favor expert-created models in these cases. This indicates that BPMNGen is capable of delivering process models with semantic accuracy on a similar level to those crafted by the expert – at least in low- to medium-complexity scenarios. Considering that these models were created fully automatically and in a fraction of the time, BPMNGen offers a substantial advantage in terms of speed and accessibility.

Only with increasing process complexity (PM4 and PM5) the participants showed a statistically significant preference for expert-created models (p = 0.003 and $p < 0.001$, respectively). This is consistent with expectations, as highly complex process descriptions often rely on implicit domain knowledge and nuanced understanding, which current LLMs like those used in BPMNGen may not yet fully capture. Nevertheless, this limitation does not diminish the strong performance of BPMNGen in simpler contexts, which are common in many practical applications. These findings highlight that BPMNGen has the potential to democratize process modeling by delivering expert-level modeling quality for a large class of use cases – particularly those that are time-sensitive, routine, or involve non-technical stakeholders. In more complex scenarios, BPMNGen offers a solid starting point, with experts

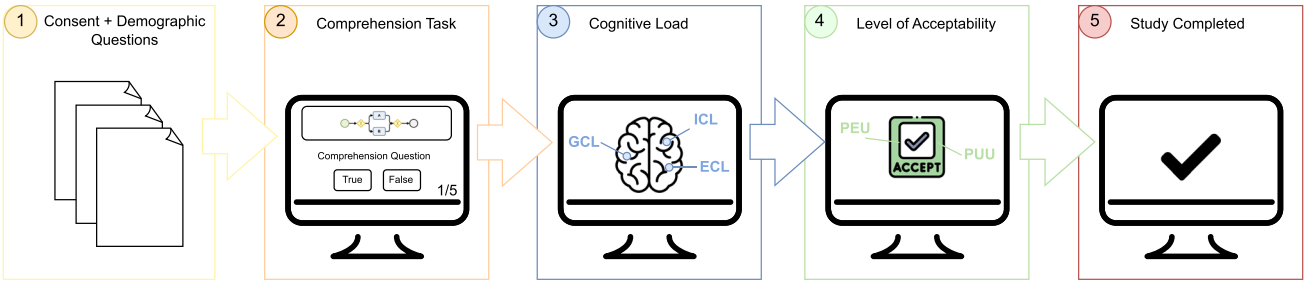


Fig. 8 Study II design

able to refine the model through prompt-based interaction with the chatbot.

Answer to Research Question RQ1: BPMNGen is capable of generating semantically accurate process models that are perceived as comparable to expert-created models in simpler and moderately complex scenarios. In these cases, participants showed no significant preference for either approach, indicating that BPMNGen can deliver expert-level modeling quality with considerably less time and effort. For more complex processes, expert-created models were favored, highlighting the importance of domain expertise in capturing subtle process semantics.

7.3 Study II: Comprehensibility of Generated Process Models

To address RQ2-RQ4, we conducted Study II, which is described in detail below, with a discussion of the results provided in Sect. 7.3.5.

7.3.1 Study Design

Study II comprises two surveys with a similar structure as Study I. In both surveys, participants first provided informed consent and answered demographic questions ① (consider Fig. 8).

In the first survey, the process models generated by BPMNGen were shown to the participants. For each process model, the participants needed to answer a true-or-false comprehension question ② (consider Fig. 8). Following this, they responded to questions assessing their cognitive load ③ (consider Fig. 8) and acceptability level ④

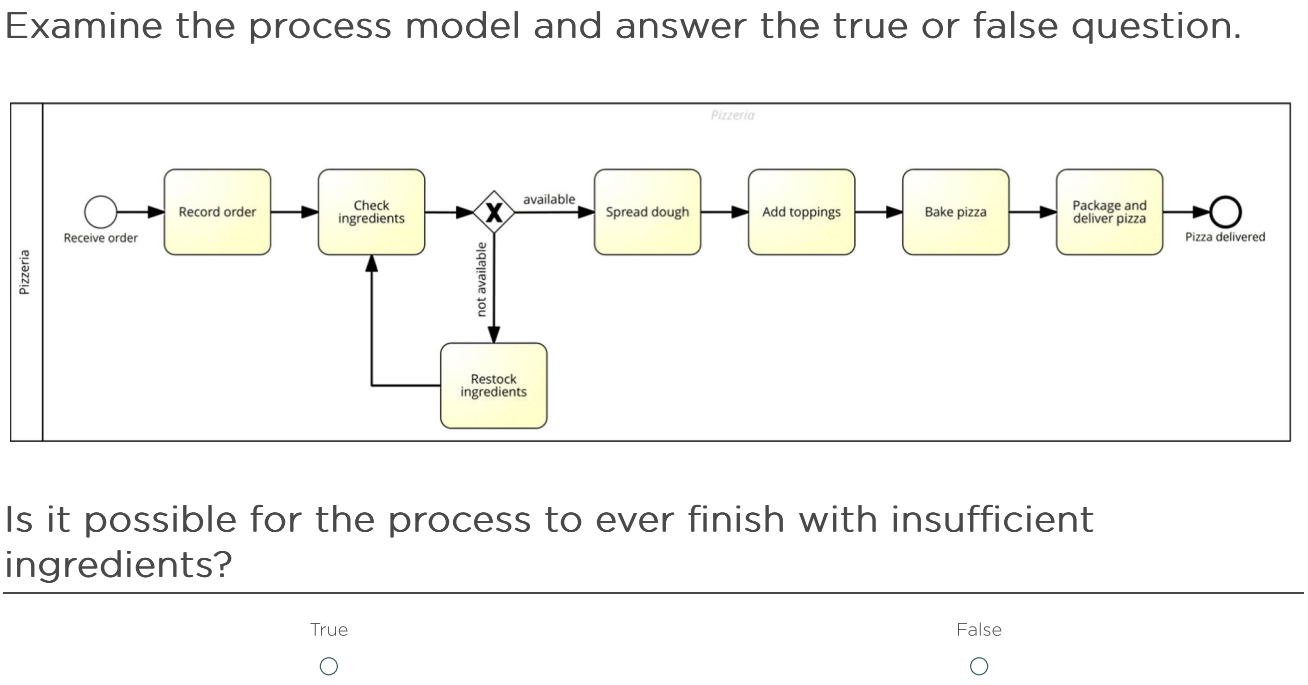


Fig. 9 Extract from Study II

Table 10 Sample description of Study IIa and IIb

Variable	BPMNGen ($N = 56$)	Expert ($N = 40$)	p-value
Gender N (%)			
female	25 (44.6)	11 (27.5)	0.066[a]
male	31 (55.4)	29 (72.5)	
other	–	–	
Age N (%)			
$<$ 25 years	54 (96.4)	26 (65.0)	<0.001[b]
$\geq$ 25 years	2 (3.6)	14 (35.0)	
Education Level N (%)			
$<$ High school diploma	1 (1.8)	2 (5.0)	
High school diploma	54 (96.4)	16 (40.0)	<0.001[b]
Bachelor Degree	1 (1.8)	17 (42.5)	
Master Degree	–	5 (12.5)	
Doctor of Philosophy (PhD)	–	–	
Experience N (%)			
no experience	2 (3.6)	18 (45.0)	
novice	43 (76.8)	12 (30.0)	<0.001[b]
intermediate	11 (19.6)	8 (20.0)	
advanced	–	2 (5.0)	
expert	–	–	

[a] Fisher's exact test

[b] Pearson-Chi-Quadrat

(consider Fig. 8) of the model. This procedure was repeated for all BPMNGen-generated process models, before completing the survey ⑤ (cf. Fig. 8). An extract of Study II is shown in Fig. 9. The second survey followed the same procedure. This time, however, the process models manually created by the human expert were presented to the participants. This design allowed for a direct comparison of the participants' responses to BPMNGen-generated models versus expert-created models. Figure 8 shows the design of Study II.

7.3.2 Study Participants

Study II, which addressed RQ2, RQ3, and RQ4, comprises two parts: Study IIa and Study IIb. Study IIa involved $N = 56$ participants, 31 male and 25 female, with an average age of 19.79 years (median = 19.0); 54 participants had already been experienced with BPMN 2.0. Study IIb involved $N = 40$ participants, of which 29 were male and 11 female. The average age of the participants was 23.83 (median = 23.0). 22 of them had already been experienced with BPMN 2.0. Table 10 presents these information in more detail as well as the p-value, representing the similarity of both groups.

7.3.3 Performance Measures

Study II examined three key aspects: Cognitive Load, Level of Acceptability, and Performance.

7.3.3.1 Cognitive Load

Cognitive load refers to the mental effort required by individuals to understand a given task or system as well as to work with it. In the context of Study II, cognitive load is evaluated as the degree to which participants comprehend and interpret the process models. Cognitive load can be further categorized into three dimensions (Sweller 1994; Paas et al. 2003):

- Intrinsic cognitive load (ICL), which is determined by the complexity of the task itself and the inherent difficulty of the material.
- Extraneous cognitive load (ECL), which arises from the way information is presented or external factors that may hinder understanding.
- Germane cognitive load (GCL), which refers to the mental effort invested in processing and understanding the material in a meaningful way.

By examining these dimensions, the study provides a comprehensive analysis of the factors influencing cognitive efforts of study participants during the comprehension process. Addressing RQ2, the cognitive load is measured using a 7-point Likert scale ranging from 1 (i.e., 'strongly disagree') to 7 (i.e., 'strongly agree').

7.3.3.2 Level of Acceptability

The Level of Acceptability is composed of two dimensions, PUU and PEU, which are derived from TAM (King and He 2006):

- *Perceived Usefulness for Understandability (PUU).* PUU refers to the extent to which individuals believe

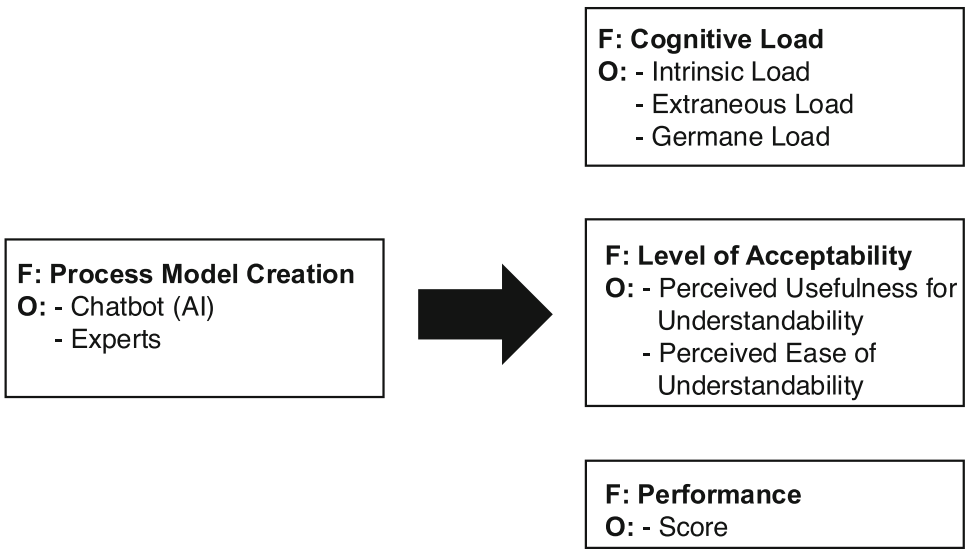


Fig. 10 Research model for Study II

that a system or output improves their ability to understand a specific concept or task. This evaluation provides insights into whether BPMNGen-generated models are perceived as more useful for understanding compared to those created by the expert. Addressing RQ3, PUU is measured using a 7-point Likert scale ranging from 'strongly disagree' (i.e., 1) to 'strongly agree' (i.e., 7).

- *Perceived Ease of Understandability (PEU).* PEU refers to the extent to which individuals believe that a system or output is easy to understand, not requiring significant mental effort. This concept is essential for evaluating the accessibility of BPMNGen-generated models compared to the process models created by the expert, focusing on how intuitive the models are. Addressing RQ3, the PEU is measured using a 7-point Likert scale ranging from 'strongly disagree' (i.e., 1) to 'strongly agree' (i.e., 7).

Table 11 Descriptive results for RQ2

Variable	BPMNGen ($N = 56$)	Expert ($N = 40$)
ICL PM1	**1.76** (0.83)	1.85 (1.08)
ICL PM2	**2.23** (1.17)	2.24 (1.08)
ICL PM3	**2.71** (1.13)	3.01 (1.19)
ICL PM4	**2.44** (1.10)	2.58 (1.14)
ICL PM5	**3.17** (1.21)	3.48 (1.41)
ECL PM1	**1.71** (0.91)	1.80 (1.26)
ECL PM2	2.04 (1.04)	**2.00** (0.96)
ECL PM3	**2.32** (1.05)	2.63 (1.27)
ECL PM4	**2.25** (1.08)	2.30 (1.02)
ECL PM5	**3.09** (1.30)	3.65 (1.51)
GCL PM1	2.04 (1.09)	**1.70** (1.12)
GCL PM2	2.18 (1.21)	**1.78** (0.87)
GCL PM3	**2.40** (1.17)	2.54 (1.25)
GCL PM4	2.42 (1.21)	**2.31** (1.25)
GCL PM5	**3.04** (1.32)	3.19 (1.59)

7.3.3.3 Performance

For each process model, the participants answered a true-or-false comprehension question, earning one point per correct answer. The sum of these points represents the overall score. The questions were not only designed to assess the comprehension of the process models, but also to motivate participants to thoroughly engage with the models. Fig. 10 shows the research model of Study II. The main theoretical factor (F) is process model creation by either BPMNGen or a human expert. The model includes three dependent theoretical factors: cognitive load, level of acceptability, and performance. Each of these factors is operationalized (O) through specific indicators – for

Table 12 Descriptive results for RQ3

Variable	BPMNGen ($N = 56$)	Expert ($N = 40$)
PUU PM1	2.70 (1.78)	**5.95** (1.26)
PUU PM2	2.86 (1.55)	**5.85** (1.14)
PUU PM3	3.02 (1.31)	**5.05** (1.50)
PUU PM4	3.00 (1.35)	**5.43** (1.41)
PUU PM5	3.82 (1.15)	**4.25** (1.41)
PEU PM1	5.90 (0.88)	**6.26** (0.83)
PEU PM2	5.78 (0.97)	**5.94** (1.17)
PEU PM3	**5.55** (0.79)	5.35 (1.23)
PEU PM4	5.56 (0.91)	**5.61** (1.22)
PEU PM5	4.74 (1.14)	**4.79** (1.29)

Table 13 Descriptive results for RQ4

Variable	BPMNGen ($N = 56$)	Expert ($N = 40$)
Score PM1	**91.1** %	90.0 %
Score PM2	**87.5** %	82.5 %
Score PM3	**94.6** %	80.0 %
Score PM4	**85.7** %	55.0 %
Score PM5	87.5 %	**90.0** %

example, cognitive load is measured via intrinsic, extraneous, and germane load; level of acceptability through perceived usefulness and ease of understanding; and performance through comprehension scores.

7.3.4 Results of Study II

This section presents the descriptive as well as inferential statistics of the results we obtained in the context of Study II.

7.3.4.1 Descriptive Results

For each of the five process models, Table 11 presents the mean and the standard deviation of the cognitive load (i.e., intrinsic, extraneous, and germane) for both groups.

The results in Table 11 indicate that the intrinsic cognitive load (ICL) and extraneous cognitive load (ECL) increase with process model complexity, with the 'Expert' models showing slightly higher mean values than the 'BPMNGen' models, particularly regarding PM4 and PM5. Germane cognitive load (GCL) follows a similar trend, with both models displaying comparable levels, though the 'Expert' model consistently scores higher GCL in the more complex scenarios (e.g., PM5). For each process model, Table 12 shows the mean and standard deviation for the level of acceptability per group.

Regarding PUU, the 'Expert' model consistently scores higher than the 'BPMNGen' model, with particularly strong differences in PM1 and PM2, indicating a higher perceived effectiveness of the 'Expert' model. Regarding PEU, both models receive relatively similar ratings, with the 'Expert' model slightly outperforming the 'BPMNGen' model except in PM3.

For each process model table 13 presents the score in both groups.

The 'BPMNGen' model demonstrates higher performance scores compared to the 'Expert' model in most scenarios, particularly regarding PM3 (94.6% vs. 80.0%) and PM4 (85.7% vs. 55.0%), indicating superior accuracy in these cases. The 'Expert' model achieves a higher score only for PM5 (90.0% vs. 87.5%), indicating its strength in specific scenarios. Overall, the results emphasize the consistency of the process models automatically generated by BPMNGen, especially for scenarios with lower complexity, whereas the models created manually by the expert reveal potential strengths with respect to more detailed contexts.

7.3.4.2 Inferential Statistics

Table 14 Inferential statistics for RQ2

	BPMNGen		Expert				
Variable	MV	SD	MV	SD	T	df	Sig. (2-Tailed)
ICL PM1	1.76	0.83	1.85	1.08	−0.466	94	0.642
ICL PM2	2.23	1.17	2.24	1.08	−0.023	94	0.982
ICL PM3	2.71	1.13	3.01	1.19	−1.248	94	0.215
ICL PM4	2.44	1.10	2.58	1.14	−0.595	94	0.554
ICL PM5	3.17	1.21	3.48	1.41	−1.136	94	0.259
ICL Sum	2.46	1.19	2.63	1.31	−1.438	403.046	0.151
ECL PM1	1.71	0.91	1.80	1.26	−0.387	94	0.700
ECL PM2	2.04	1.04	2.00	0.96	0.171	94	0.865
ECL PM3	2.32	1.05	2.63	1.27	−1.238	73.498	0.220
ECL PM4	2.25	1.08	2.30	1.02	−0.229	94	0.820
ECL PM5	3.09	1.30	3.65	1.51	−1.948	94	0.054
ECL Sum	2.29	1.17	2.48	1.37	−1.612	384.367	0.108
GCL PM1	2.04	1.09	1.70	1.12	1.474	94	0.144
GCL PM2	2.18	1.21	1.78	0.87	1.801	94	0.075
GCL PM3	2.40	1.17	2.54	1.25	−0.581	94	0.563
GCL PM4	2.42	1.21	2.31	1.25	0.421	94	0.675
GCL PM5	3.04	1.32	3.19	1.59	−0.480	94	0.632
GCL Sum	2.41	1.24	2.30	1.34	0.940	478	0.348

Table 15 Inferential statistics for RQ3

	BPMNGen		Expert				
Variable	MV	SD	MV	SD	T	df	Sig. (2-Tailed)
PUU PM1	2.70	1.78	5.95	1.26	−10.494	93.998	< 0.001*
PUU PM2	2.86	1.55	5.85	1.14	−10.334	94	< 0.001*
PUU PM3	3.02	1.31	5.05	1.50	−7.03	94	< 0.001*
PUU PM4	3.00	1.35	5.43	1.41	−8.518	94	< 0.001*
PUU PM5	3.82	1.15	4.25	1.41	−1.585	72.973	0.117
PUU Sum	3.08	1.49	5.31	1.47	−16.236	478	< 0.001*
PEU PM1	5.90	0.88	6.26	0.83	−2.023	94	0.046*
PEU PM2	5.78	0.97	5.94	1.17	−0.734	94	0.465
PEU PM3	5.55	0.79	5.35	1.23	0.922	61.730	0.360
PEU PM4	5.56	0.91	5.61	1.22	−0.219	68.395	0.827
PEU PM5	4.74	1.14	4.79	1.29	−0.186	94	0.853
PEU Sum	5.51	1.02	5.59	1.25	−0.769	372.844	0.442

Table 14 compares the scores for the instrinsic cognitive load (ICL), the extraneous cognitive load (ECL), and the germane cognitive load (GCL) between the BPMNGen and expert group. Each category includes five performance measures (PM1 to PM5) and an overall score (i.e., ICL sum, ECL sum, GCL sum). For each variable, the table provides the mean and standard deviations for both groups, alongside inferential statistics including the t-test values (T), degrees of freedom (df), and two-tailed significance levels (Sig. 2-Tailed).

To answer *RQ2*, a t-test for independent samples was conducted to assess whether there were significant differences in cognitive load assessments between the BPMNGen and expert groups. For the ICL sum, which represents the aggregated ICL scores across all five process models (PM1 to PM5), there was no significant difference ($t = -1.438$, $p = 0.151$). Note that this pattern is consistent across all other categories (ECL and GCL), where no statistically significant differences were observed for either the individual performance measures or the overall scores. To conclude, the t-test for independent samples indicates that there are no significant differences in cognitive load assessments between the BPMNGen and expert group, neither for individual process models nor for the overall scores. Thus, based on the available data, we did not find evidence that BPMNGen's process models impose a higher cognitive load than those created by the human expert.

To answer *RQ3*, a t-test for independent samples was conducted to compare the groups (cf. Table 15). Regarding PUU, significant differences were found for PM1 to PM4 and in the overall PUU Sum ($t = -16.236$, $p < 0.001$), indicating that the expert group rated these measures significantly higher than the BPMNGen group. However, no significant differences were observed for PM5, as the p-value exceeded 0.05.

Regarding PEU, PM1 showed a significant difference ($t = -2.023$, $p = 0.046$), with the expert group rating this construct again higher than the BPMNGen group. Although none of the other individual measures (PEU PM2 to PM5) showed statistically significant differences (p-values > 0.05), the same applies to the sum of PEU.

To answer *RQ4*, the inferential statistics was based on a t-test for independent samples and revealed statistically significant differences between the BPMNGen group and the expert group for several measures (cf. Table 16). Specifically, significant differences were observed for the Score of PM3 ($t = 2.066$, $p = 0.043$), where the BPMNGen group achieved higher scores compared to the expert group, the Score of PM4 ($t = 3.317$, $p = 0.001$), with the BPMNGen group performing significantly better than the

Table 16 Inferential statistics for RQ4

Variable	BPMNGen Frequency	Expert Frequency	T	df	Sig. (2-Tailed)
Score PM1	91.1 %	90.0 %	0.176	94	0.861
Score PM2	87.5 %	82.5 %	0.679	94	0.499
Score PM3	94.6 %	80.0 %	2.066	56.474	0.0043*
Score PM4	85.7 %	55.0 %	3.317	65.452	0.001*
Score PM5	87.5 %	90.0 %	−0.375	94	0.708
Score Sum	89.29%	79.5 %	2.871	356.004	0.004*

expert group, and the overall Score Sum ($t = 2.871$, $p = 0.004$), indicating that the BPMNGen group outperformed the expert group in the overall score.

In contrast, no significant differences were found between the BPMNGen group and the expert group for the scores of PM1, PM2, and PM5, as their p-values exceeded the threshold of 0.05.

In summary, the results of the t-test for independent samples have shown that the BPMNGen group significantly outperformed the expert group in the overall score (Score Sum) and specific performance measures (Score of PM3 and Score of PM4), while the remaining measures did not exhibit statistically significant differences.

7.3.5 Discussion

Study II expanded the analysis to assess the usability and comprehension of the created models, focusing on the cognitive load (RQ2), the level of acceptability (RQ3), and the comprehension performance (RQ4). Participants evaluated five BPMNGen-generated models (Study IIa) and five expert-created models (Study IIb) under controlled conditions. Across all dimensions of cognitive load – intrinsic (ICL), extraneous (ECL), and germane (GCL) – no statistically significant differences were found between the two groups. Based on our data, this result indicates that there is no evidence that the comprehension of BPMNGen-generated process models is more mentally demanding than the comprehension of process models manually created by the expert. Note that from a usability perspective, this is a highly promising finding: despite being fully automatically generated, BPMNGen's outputs were perceived as equally interpretable and cognitively manageable as those crafted through expert-driven modeling. This cognitive parity is particularly relevant in practice, where model comprehension often involves diverse stakeholders with varying levels of process modeling expertise. BPMNGen demonstrates the potential to generate process models that are not only efficient to produce but also easy to interpret, lowering the entry barrier for non-experts and enabling broader participation in modeling activities.

> **Answer to Research Question RQ2:** The cognitive load experienced during the comprehension of BPMNGen-generated process models does not differ significantly from that of expert-created models across varying levels of process complexity. Based on these results, we do not find evidence that BPMNGen's models impose a higher cognitive load. However, they require substantially less modeling effort and time.

PUU results indicate a significant advantage for expert-created models across most process models (PM1–PM4), as expert models received consistently higher ratings than those generated by BPMNGen (PUU Sum, $p < 0.001$). However, for PM5, the difference in PUU ratings is not significant ($p = 0.117$), indicating that in this specific scenario, the usefulness of BPMNGen and expert models are perceived similarly. Regarding PEU, expert-created models were rated significantly higher for PM1 ($p = 0.046$), indicating that participants found them easier to understand in this particular scenario. However, no significant differences were observed for PEU in the other four models and in the overall PEU Sum ($p = 0.442$), indicating that BPMNGen- and expert-created models seem to be comparable regarding their accessibility and ease of understanding.

> **Answer to Research Question RQ3:** The level of acceptability of BPMNGen-generated models varies with process complexity. Expert-created models were perceived being significantly more useful (PUU) in simpler and moderately complex scenarios (PM1–PM4), whereas no significant difference was observed for PM5 – the most complex scenario. This indicates that the usefulness of BPMNGen improves relatively to expert models with increasing complexity. Concerning ease of understanding (PEU), expert models were only rated significantly higher in the simplest scenario (PM1); for all other models, there was no significant difference. Thus, BPMNGen-generated models are generally considered being equally understandable, and increasingly useful as process complexity grows.

Interestingly, for RQ4, BPMNGen-generated models outperformed expert-created models in the context of specific tasks, such as PM3 ($p = 0.043$) and PM4 ($p = 0.001$). These findings indicate that BPMNGen-generated models might offer advantages in scenarios for which a clear and concise process logic is sufficient for model understanding. However, expert models outperformed BPMNGen models in the context of PM5, a scenario involving more elaborate decision-making processes. This disparity highlights the potential of LLM-based conversational frameworks for routine tasks while emphasizing the enduring value of expert knowledge for complex and nuanced scenarios.

> **Answer to Research Question RQ4:** Comprehension performance varied depending on process complexity. BPMNGen-generated models significantly outperformed expert-created models in moderately complex scenarios (PM3 and PM4), suggesting that their clear and structured representation supports effective understanding in such tasks. In contrast, expert models led to better comprehension in the most complex scenario (PM5), indicating their

strength in capturing nuanced and decision-heavy processes.

Discussion. The models automatically generated by BPMNGen demonstrate strong potential, especially in scenarios with lower to moderate complexity. These results are particularly encouraging given the substantial time and resource savings offered by fully automated modeling. As BPMNGen continuously evolves, further improvements in model quality and semantic precision can be expected. Our ongoing development efforts focus on expanding the modeling capabilities – for instance, by integrating more complex BPMN 2.0 elements (e.g., data and data flows, modularization through sub-processes) and choreography diagrams – to support more advanced process scenarios. In addition, we plan to enhance BPMNGen with Retrieval-Augmented Generation (RAG). This could enrich the modeling process by allowing BPMNGen to incorporate external contextual information – such as domain-specific documentation, information from process repositories, or organizational guidelines – into its responses. By grounding model generation in relevant external knowledge, RAG approaches may help us to improve the semantic quality and correctness of the resulting BPMN models, particularly in regulated or knowledge-intensive domains where precision and domain alignment are essential. Furthermore, we are actively refining our prompting strategies to enhance the accuracy, expressiveness, and practical applicability of semi-automatically generated models. These continuous improvements strengthen BPMNGen's position as a scalable and accessible assistant for process modeling in diverse real-world contexts.

8 Limitations and Threats to Validity

While the evaluation studies of BPMNGen offer valuable insights into the capabilities of LLM-based process modeling, several limitations and threats to validity need to be acknowledged. These pertain both to the design and execution of the studies as well as the generalizability and robustness of the findings.

One limitation lies in the simplicity of the process scenarios used in both studies. Scenarios such as baking a pizza or requesting vacation were deliberately chosen to match the current capabilities of BPMNGen and to ensure accessibility for participants with limited modeling expertise. While this improves ecological validity in everyday contexts, it restricts conclusions about the applicability of BPMNGen to more complex, technical, or domain-specific processes. Consequently, future work should assess performance in more sophisticated settings that reflect real-world industry requirements.

Another concern involves the composition of the participant samples. Although a range of BPMN experience levels was represented, most participants were university students. This raises questions about the transferability of results to professional environments with trained modelers or domain experts. Although efforts were made to balance groups demographically and reduce selection bias through random assignment (in Study II) and within-subject comparisons (in Study I), residual group differences – such as age or familiarity with modeling tools – may still influence the results.

To ensure semantic comparability, the models used in both studies were closely aligned with the original textual descriptions, and visual adjustments (e.g., straightening arrows or repositioning flow elements) were applied consistently without altering model semantics. Nevertheless, subtle layout features or perceived visual quality may have unconsciously biased participant evaluations.

The measurement of constructs such as perceived usefulness, ease of understanding, and cognitive load relied on validated Likert-scale instruments grounded in established theories (e.g., Technology Acceptance Model, Cognitive Load Theory). However, self-report instruments remain vulnerable to social desirability and subjective interpretation. While comprehension was also evaluated using true-or-false questions to introduce an objective dimension, these questions – by their binary nature – capture only a fraction of the depth of understanding. Incorporating behavioral or physiological metrics such as task completion time, error rates, or eye tracking in future studies would provide a more comprehensive assessment of model comprehensibility and user experience. Moreover, BPMNGen-generated models tend to contain fewer elements on average than expert-created counterparts. This structural compactness aligns with BPMNGen's goal of generating accessible models, particularly for non-expert users. The study included process scenarios with increasing difficulty to account for model complexity, but it did not separately analyze the exact number of modeling elements. It is therefore possible that the reduced number of elements contributed to the lower cognitive load and higher perceived ease of understanding observed for BPMNGen models. Future research could isolate this factor more directly to further disentangle the impact of model size from other dimensions of model quality.

A particular risk lies in the evolving nature of large language models themselves. As LLMs continue to develop rapidly, many aspects – such as model architecture, token limitations, memory handling, or API behavior – are subject to frequent change. This dynamic evolution introduces a degree of variability and hinders reproducibility over time. Moreover, improvements in prompt engineering, fine-tuning strategies, or tool integration may

soon alter the effectiveness and output quality of frameworks like BPMNGen. While our findings reflect the state of the technology at the time of the study, future research must account for such technological shifts to ensure continued relevance.

Another threat arises from the dependence of our study on a specific underlying LLM. The process models generated in our study were based on GPT-3.5 Turbo, and different LLMs may produce process models that differ in structure, level of detail, or modeling style. While BPMNGen is designed to seamlessly adopt newer LLM versions, such changes can lead to variations in the generated models and thus affect the stability of the study's results. Consequently, the generalizability of our findings to other LLM configurations must be interpreted with caution.

Another threat concerns the dependence of our results on the specific textual input scenarios used in the study. All BPMNGen process models were generated from fixed, author-defined process descriptions using a consistent zero-shot prompting setup. While this ensures comparability across scenarios, LLMs are known to be sensitive to variations in wording and structure, meaning that different formulations of the same process description could lead to different outputs. Consequently, our findings apply to the specific input formulations used and should not be generalized to arbitrary natural-language descriptions.

A further threat relates to the limited coverage of BPMN 2.0 elements evaluated in this work. At the time of the study, BPMNGen supported only the basic BPMN 2.0 elements needed for the selected scenarios (e.g., activities, events, gateways, sequence flows, pools, and lanes). More advanced elements such as message flows, collaborations, or subprocesses were not yet reliably supported and therefore not included. Accordingly, the evaluation focuses on a constrained subset of BPMN 2.0 elements for which BPMNGen provided trustworthy and reproducible results. Consequently, the expressiveness of BPMNGen cannot be generalized to the full BPMN 2.0 specification.

From a statistical perspective, repeated measures across multiple scenarios strengthened the reliability of findings, but the relatively small sample sizes and group-specific designs still limit generalizability. A more robust experimental setup could involve counterbalanced within-subject designs or crossover studies to control for individual differences and order effects more effectively.

In summary, while the presented findings indicate promising capabilities for LLM-supported process modeling, they must be interpreted within the methodological and technical boundaries outlined above. Addressing these limitations in future research will be critical for realizing the full potential of BPMNGen in diverse, real-world modeling contexts.

9 Conclusions and Outlook

This article introduced BPMNGen, an LLM-based conversational framework for automatically generating BPMN 2.0 process models from natural language descriptions. BPMNGen combines modern natural language processing techniques with large language models to enable users to create and iteratively refine process models via a sophisticated interface. The BPMNGen architecture follows a modular design, consisting of five key components which enables scalable and efficient process model generation with seamless user interaction.

To evaluate the capabilities of BPMNGen-generated models, we conducted two empirical studies. These investigated four research questions focusing on model quality, model comprehensibility, cognitive load in the context of model reading, and perceived user acceptance.

Regarding semantic accuracy (RQ1), our findings have shown that participants exhibited no clear preference between BPMNGen- and expert-generated models in the context of simple process descriptions. However, for more complex process scenarios, participants favored expert-generated models, indicating that the current BPMNGen capabilities are still limited when handling intricate semantics or structural depth. Nevertheless, for less complex modeling tasks, BPMNGen-generated models can be considered a valid and efficient alternative. Note that we are continuously advancing BPMNGen to tackle its current limitations and make it more suitable for generating complex scenarios as well.

Concerning cognitive load (RQ2), no significant differences were found between BPMNGen- and expert-generated models across the three cognitive dimensions: intrinsic, extraneous, and germane. This indicates that both model types impose comparable mental effort, and that the complexity of the modeled process plays a more decisive role in shaping cognitive demand than the source of the model.

Regarding *Perceived Usefulness* (RQ3), participants rated expert-generated models significantly better in most cases, particularly in simple modeling scenarios. Regarding *Ease of Understandability* only PM1 was rated significantly better. Although BPMNGen-generated models performed similarly in complex settings with respect to ease of understanding, their perceived helpfulness was generally lower. These results emphasize the potential value of hybrid approaches, where conversational frameworks serve as initial model generators with the automatically generated models then being iteratively refined by human experts.

Finally, comprehension performance (RQ4) was found to be task-dependent. While BPMNGen-generated process models led to better comprehension outcomes for simple

processes, expert models were more effective in conveying complex logic. This indicates that the simplicity of models generated with BPMNGen can facilitate quick understanding in simpler scenarios, but that greater sophistication is needed to ensure comprehension in more complex cases. Our findings have revealed several promising directions for future research. One avenue worth exploring is the extension of BPMNGen's modeling capabilities to better support advanced BPMN 2.0 constructs, such as data objects, artifacts, boundary events, sub-processes, and other advanced BPMN 2.0 elements. Investigating how these features can be reliably generated and integrated in a conversational setting would contribute to increase the functional depth and practical applicability of the framework.

Further research may investigate multimodal inputs, such as hand-drawn sketches, speech, or structured templates, enabling more flexible user interaction. Similarly, process model modularization – the ability to break down large processes into reusable sub-processes – could improve scalability and maintainability in large modeling projects.

Moreover, adopting a *model-to-text* approach – i.e., translating BPMN process models back into natural language explanations – may offer significant benefits for onboarding, documentation, and compliance. Such functionality could help make process logic more transparent and accessible, particularly for non-technical stakeholders. By automatically generating human-readable descriptions from formal models, BPMNGen could contribute to closing the communication gap between model creators and process participants, thereby improving process understanding and alignment across the organization.

Beyond BPMN 2.0, BPMNGen could be extended to support alternative modeling paradigms, such as object-centric process modeling (Künzle and Reichert 2010; Steinau et al. 2019; van der Aalst 2019, 2021) or declarative approaches (Reijers et al. 2013; De Giacomo et al. 2015). Exploring these formats may broaden the applicability of conversational modeling to use cases where traditional procedural models are less suitable.

In conclusion, BPMNGen represents a promising step toward making business process modeling more accessible through conversational AI. While it already shows strong potential in straightforward modeling scenarios, there are numerous opportunities to expand its scope and effectiveness. The insights and evaluations presented in this work provide a robust foundation for further research in this emerging field.

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